

Module 08: Sliding Window Protocols: Go-Back-N ARQ Architecture

Unit 2: Data Link Layer & MAC Sublayer

Pipelined ARQ Protocols

Deep mathematical dive into Go-Back-N ARQ: sliding window pointer mechanics, cumulative acknowledgments, formal proof for window sizing $W_S = 2^m - 1$, and numerical derivations for 100% efficiency.

Sliding Window Protocols: Go-Back-N (GBN) ARQ Architecture & Window Mechanics

1. Window Pointers & Buffer Mechanics

Sender Window ($m = 3 \text{ bits} \rightarrow W_S = 2^3 - 1 = 7$):



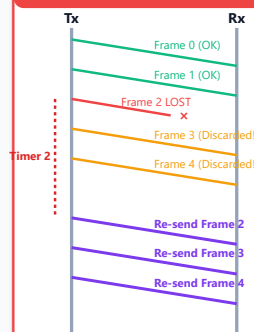
Receiver Window (Strictly $W_R = 1$):



Receiver has NO BUFFER for out-of-order frames!

If Frame 2 is lost, arriving Frames 3, 4, 5 are DISCARDED unconditionally!

2. Error Recovery: Frame Loss & Retransmit-N



The Go-Back-N Penalty:

Even though Frames 3 & 4 arrived without noise, receiver rejected them because Frame 2 was missing!

Sender must GO BACK and re-send ALL N outstanding frames!

Wastes enormous bandwidth on noisy high-speed links.

Mathematical Foundation: Window Size Proof ($W_S < 2^m$) & Pipelined Efficiency

Why $W_S = 2^m - 1$?

If $W_S = 2^m$ (e.g., $m=2$, $W_S=4$, sequence {0,1,2,3}). If sender sends frames 0,1,2,3 and all ACKs are lost, receiver expects Frame 0 of NEW round. When sender retransmits old Frame 0, receiver CANNOT tell if it's new or old!

Efficiency Formula:

$\eta = W_S / (1 + 2a)$ for $W_S < (1 + 2a)$ | $\eta = 100\%$ for $W_S \geq (1 + 2a)$

Optimal Window:

To achieve 100% utilization: Window Size $W_S \geq 1 + 2a = 1 + 2(Tp/Tt)$. Pipelining fills the transmission pipe!

1. Pipelining & Go-Back-N Mechanics

Pipelining overcomes the single-frame waiting penalty of Stop-and-Wait by maintaining a window of outstanding unacknowledged frames (W_S). Receiver window is strictly $W_R = 1$, enforcing sequential in-order acceptance without out-of-order buffering.

2. Formal Proof of $W_S = 2^m - 1$

If $W_S = 2^m$, a complete loss of an entire burst of ACKs forces retransmissions of old frames that wrap around modulo 2^m , making it mathematically impossible for the receiver to distinguish duplicate retransmitted frames from brand new frames of the next cycle.

Go-Back-N Mathematical Governing Equations

Sender Window: $W_S = 2^m - 1$ | Receiver Window: $W_R = 1$

Efficiency: $\eta = W_S / (1 + 2a)$ [for $W_S < 1 + 2a$]

Optimal Window for 100% η : $W_S \geq 1 + 2a$